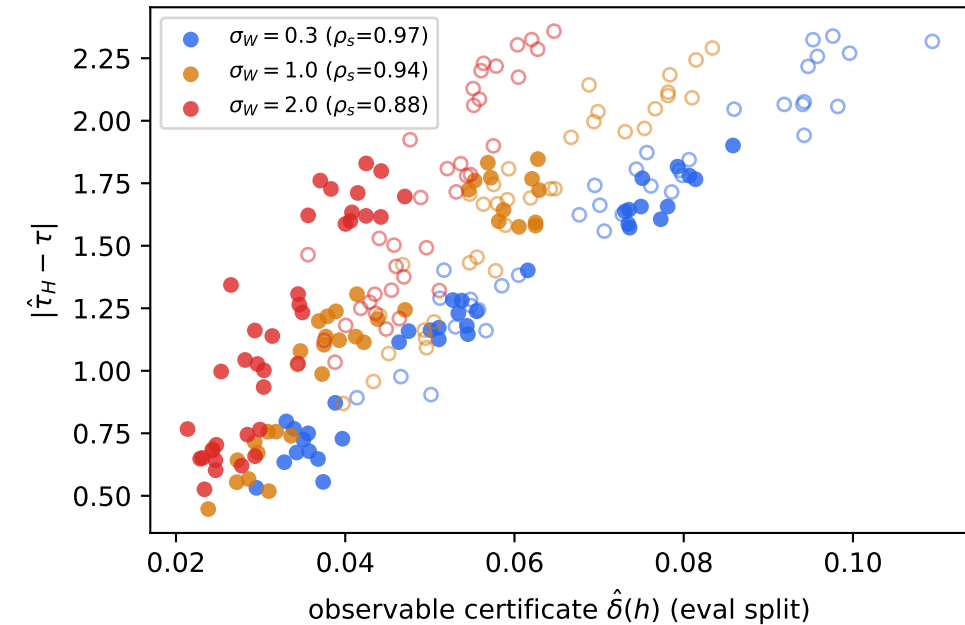
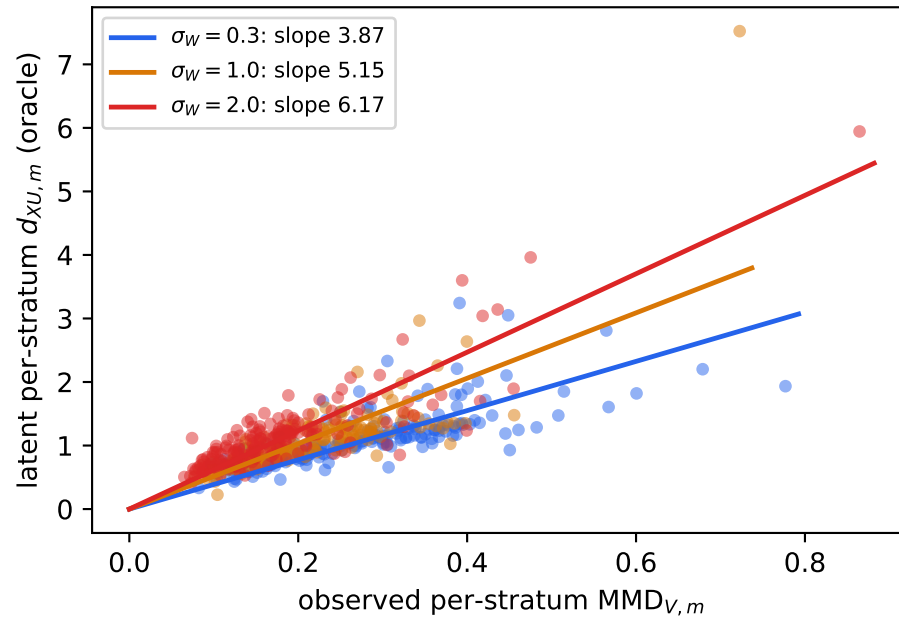


Chain validation: small observable discrepancy  $\Rightarrow$  small latent imbalance  $\Rightarrow$  small bias (pooled over  $\gamma_U \in 0.5, 1.0, 2.0$ )

(a) certificate vs realized bias  
(filled: learned  $\hat{h}$ ; hollow: random  $h$ )



(b) channel inversion: slope  $\approx$  empirical  $\omega_K$



(c) latent imbalance bounds bias

